Coal Tar and Coal Tar Pitch: Human health tier II assessment

13 February 2015

Chemicals in this assessment

Chemical Name in the Inventory	CAS Number
Tar, coal	8007-45-2
Tar, coal, high temperature	65996-89-6
Pitch, coal tar, high temperature	65996-93-2
Pitch, coal tar, petroleum	68187-57-5
Pitch, coal tar, low temperature	90669-57-1
Pitch, coal tar, low temperature, heat treated	90669-58-2

Preface

This assessment was carried out by staff of the National Industrial Chemicals Notification and Assessment Scheme (NICNAS) using the Inventory Multi-tiered Assessment and Prioritisation (IMAP) framework.

The IMAP framework addresses the human health and environmental impacts of previously unassessed industrial chemicals listed on the Australian Inventory of Chemical Substances (the Inventory).

The framework was developed with significant input from stakeholders and provides a more rapid, flexible and transparent approach for the assessment of chemicals listed on the Inventory.

Stage One of the implementation of this framework, which lasted four years from 1 July 2012, examined 3000 chemicals meeting characteristics identified by stakeholders as needing priority assessment. This included chemicals for which NICNAS already held exposure information, chemicals identified as a concern or for which regulatory action had been taken overseas, and chemicals detected in international studies analysing chemicals present in babies' umbilical cord blood.

Stage Two of IMAP began in July 2016. We are continuing to assess chemicals on the Inventory, including chemicals identified as a concern for which action has been taken overseas and chemicals that can be rapidly identified and assessed by using Stage One information. We are also continuing to publish information for chemicals on the Inventory that pose a low risk to human health or the environment or both. This work provides efficiencies and enables us to identify higher risk chemicals requiring assessment.

The IMAP framework is a science and risk-based model designed to align the assessment effort with the human health and environmental impacts of chemicals. It has three tiers of assessment, with the assessment effort increasing with each tier. The Tier I assessment is a high throughput approach using tabulated electronic data. The Tier II assessment is an evaluation of risk on a substance-by-substance or chemical category-by-category basis. Tier III assessments are conducted to address specific concerns that could not be resolved during the Tier II assessment.

These assessments are carried out by staff employed by the Australian Government Department of Health and the Australian Government Department of the Environment and Energy. The human health and environment risk assessments are conducted and published separately, using information available at the time, and may be undertaken at different tiers.

This chemical or group of chemicals are being assessed at Tier II because the Tier I assessment indicated that it needed further investigation.

For more detail on this program please visit:**www.nicnas.gov.au**

Disclaimer

ACRONYMS & ABBREVIATIONS

Grouping Rationale

The chemicals in this group are coal tars and coal tar pitches. Coal tars are by-products of the destructive distillation (or carbonisation) of coal when producing coke and/or coal gas. Coal tar pitch is the residue from coal-tar distillation.

The chemicals are UVCBs - substances that are of unknown or variable composition, complex reaction products and biological materials. Coal tars are a complex combination of polycyclic aromatic hydrocarbons (PAHs), phenols, heterocyclic oxygen, sulphur and nitrogen compounds. Coal tar pitches consist of four to seven membered aromatic hydrocarbon ring systems as well as their methylated and polymethylated derivatives, mono- and polyhydroxylated derivatives, and heterocyclic compounds.

The composition of the coal tars (and coal tars pitch derived thereof) is process dependent, with the major variable being the PAHs. The composition and properties of a coal tar depend primarily on the temperature of the carbonisation and, to a lesser extent, on the nature (source) of the coal used as feedstock. The content of PAHs in coal tars increases as the carbonisation temperature increases. PAH concentrations in coal-tar products can range from less than 1 % up to 70 % or more. Benzo[a]pyrene levels have been reported up to 6.4 g/kg in coal tar and up to 13.2 g/kg for high temperature coal tar pitch, with similar or higher levels of other PAHs reported (IARC, 1985; ATSDR, 2002).

Import, Manufacture and Use

Australian

The chemical, high temperature coal tar pitch (CAS No. 65996-93-2) is listed on the 2006 High Volume Industrial Chemicals List (HVICL) with reported commercial use as a binding and conductive agent and a total reported volume of 10000–99999 tonnes.

The non-industrial uses identified for coal tar in Australia as an ingredient in therapeutic goods including lotions, creams and scalp cleansers are listed on the Australian Register of Therapeutic Goods (ARTG).

International

The following international uses have been identified through:

- the European Union (EU) Registration, Evaluation and Authorisation of Chemicals (REACH) dossiers;
- Galleria Chemica;
- the Substances and Preparations in the Nordic countries (SPIN) database;
- the European Commission Cosmetic Ingredients and Substances (CosIng) database;
- the United States (US) Personal Care Product Council International Nomenclature of Cosmetic Ingredients (INCI) Dictionary;
- the Organisation for Economic Co-operation and Development High Production Volume chemical program (OECD HPV);
- the US Environmental Protection Agency's Aggregated Computer Toxicology Resource (ACToR);
- the US National Library of Medicine's Hazardous Substances Data Bank (HSDB); and
- various international assessments (IARC, 1985; ASTDR, 2002; IARC, 2012; NTP, 2014).

Coal tars are reported to have site-limited use in producing coal-tar products and as fuel. Commercial uses in surface coatings have also been reported. Coal tars are listed in the US Personal Care Product Council INCI Dictionary with the reported functions of antidandruff agents; cosmetic biocides and denaturants. Coal tars also have reported use in hair care products. Overall frequency of coal tar use was reported to be low (three products) (Personal Care Products Council, 2011). Coal tar preparations have been used for many years in therapeutic products (non-industrial) for skin conditions.

Coal tar pitches have reported site-limited use in manufacturing other chemicals and commercial uses including:

- as a binder for electrodes;
- in road paving;
- in roofing construction; and
- in surface coatings such as pipe-coating enamels and protective coatings for steel works.

The chemical, CAS No. 65996-93-2, was listed as an ingredient in three driveway sealant products (concentration unspecified) in the USA. Coal tar pitch was listed as ingredient in one sealant product and two driveway sealer products (up to a concentration of 45 %) in the USA. However, the CAS No. used for this chemical is related to coal tar (CAS No. 8007-45-2); therefore, the exact identity of the ingredient is uncertain (Household Products Database, US Department of Health and Human Services).

Restrictions

Australian

These chemicals are listed in the *Poisons Standard—the Standard for the Uniform Scheduling of Medicines and Poisons* (SUSMP) in Appendix C (SUSMP, 2014) as the following:

'COAL TAR for cosmetic use other than in therapeutic goods.'

Coal tar pitch products, being derivatives of coal tar, would be captured by this entry.

International

The chemicals are listed on the following (Galleria Chemica):

- Association of South Eastern Asian Nations (ASEAN) Cosmetic Directive Annex II Part 1—List of substances which must not form part of the composition of cosmetic products.
- EU Cosmetics Regulation 1223/2009 Annex II—List of substances prohibited in cosmetic products; and
- New Zealand Cosmetic Products Group Standard—Schedule 4: Components cosmetic products must not contain.

CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 are only listed if the chemical contains >0.005 % w/w benzo[a]pyrene.

The chemicals as 'Coal tars crude and refined' are listed in Health Canada List of prohibited and restricted cosmetic ingredients (The Cosmetic Ingredient 'Hotlist').

The chemicals are restricted under Annex XVII to REACH Regulations. 'The chemical cannot be used in substances and preparations placed on the market for sale to the general public in individual concentrations ≥0.1% ' (European Parliament and Council 1999; European Parliament and Council 2006; European Parliament and Council 2008). CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 are only listed if the chemical contains >0.005 % w/w benzo[a]pyrene.

Existing Worker Health and Safety Controls

Hazard Classification

The chemicals in the group are classified as hazardous, with the following risk phrase for human health in the Hazardous Substances Information System (HSIS) (Safe Work Australia):

- R45 Carc. Cat 1 (carcinogenicity) (CAS Nos. 8007-45-2 and 65996-89-6); and
- R45 Carc. Cat 2 (carcinogenicity) (CAS Nos. 65996-93-2, 68187-57-5, 90669-57-1 and 90669-58-2).

These classifications are subject to note H.

'Note H: The classification and label shown for this substance applies to the dangerous property(ies) indicated by the Risk Phrase(s) in combination with the category(ies) of danger shown. The manufacturers, distributors and importers of this substance shall be obliged to carry out an investigation to make themselves aware of the relevant and accessible data which exists for all other properties to classify and label the substance.'

The classifications for CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 are subject to note M.

'Note M: The classification as a carcinogen need not apply if it can be shown that the substance contains less than 0.005% w/w benzo[a]pyrene (EINECS no. 200-028-5). This note only applies to certain complex coal-derived substances in Annex I.'

Exposure Standards

Australian

The chemical (CAS No. 65996-93-2) has an exposure standard of 0.2 mg/m^3 time weighted average (TWA).

International

The following exposure standards are identified for coal tar pitch volatiles (Galleria Chemica).

An exposure limit of 0.14–0.2 mg/m^3 (TWA) in different countries such as the Belgium, Bulgaria, Canada, China, Iceland, Ireland, Italy, Korea, Malaysia, New Zealand, Singapore, South Africa, the United Arab Emirates, the USA (Idaho, Vermont) and Venezuela. An exposure limit of 0.02 mg/m^3 TWA was noted for Mexico and Nicaragua. A 0.03 mg/m^3 short-term exposure limit (STEL) was noted for Mexico.

The American Conference of Governmental Industrial Hygienists (ACGIH) recommends a threshold limit value (TLV) of 0.2 mg/m^3) TWA (as benzene solubles). 'This value is intended to minimize the potential for an increase in the incidence of lung and other tumors.' (ACGIH, 2011).

Health Hazard Information

When data specific to individual chemicals have been used for read across purposes, the CAS No. has been identified. Where the group as a whole is being referred to, the generic term 'coal tar' or coal tar pitches' has been used. Data for polycyclic aromatic hydrocarbons are included as supporting data where appropriate.

Toxicokinetics

Limited data are available. Toxicological data indicate that the chemicals are absorbed through all routes of exposure.

Acute Toxicity

Oral

Data available for coal tar and coal tar pitch indicate that the chemicals have low acute oral toxicity. The chemicals (CAS Nos. 65996-89-6 and 65996-93-2) have low acute toxicity based on results from animal tests following oral exposure. The median lethal doses (LD50s) for CAS Nos. 65996-89-6 and 65996-93-2 in rats are >2000 mg/kg bw and >15000 mg/kg bw, respectively. Observed sub-lethal effects included piloerection, apathy, decreased response to stimuli, hunched posture and ischaemic mucous membranes (REACHa; REACHb).

Dermal

Data available indicate that the chemicals have low acute dermal toxicity. The chemical (CAS No. 65996-93-2) has low acute toxicity based on results from animal tests following dermal exposure. The LD50 in rats is >2000 mg/kg bw. Observed sub-lethal effects included stained fur and treated skin area (REACHb).

Inhalation

No data are available.

Corrosion / Irritation

Respiratory Irritation

Evidence for respiratory irritation has been observed in animals and humans following repeated exposures (refer **Repeated dose toxicity** section).

Skin Irritation

Studies were performed in accordance with OECD Test Guideline (TG) 404. The chemical (CAS No. 65996-93-2), applied to intact and scarified New Zealand White rabbit skin, produced no erythema or oedema after 72 hours. The chemical was found to be a non irritant (REACHb).

Eye Irritation

The chemical, CAS No. 65996-93-2, was reported as not irritating to the eyes when tested according to OECD TG 405. The average scores for conjunctival redness and conjunctival chemosis were both given as 2. The effects were reversible within 24 hours after application (REACHb).

Eye irritant effects have been reported in mice and rabbits for coal tar products (ATSDR, 2002), although there are insufficient data to make a recommendation for classification.

Observation in humans

Evidence of skin and eye irritation in humans following exposure to coal tar pitches has been reported (ATSDR, 2002).

Reddened eyelids and conjunctivae have been reported following acute exposures to pitch fumes; chronic exposures led to corneal damage.

Exposure to ultraviolet (UV) has been reported to increase irritant responses, although more data are available for coal tar distillates (ATSDR, 2002; REACH).

In a non-guideline study to determine the minimal dose required to induce a phototoxic exposure, coal tar (CAS No. 8007-45-2) was applied to intact, untanned human skin (REACHa). The skin was exposed to the chemical for 15, 30, 60, 90, 120 or 180 minutes and then exposed to UV A light for 4–60 minutes. Phototoxicity was noted in all subjects and was reported as fully reversible after 30 hours (REACHa).

The phototoxic effects of several PAHs were compared by treating human fibroblasts with these PAHs and then irradiating them with ultraviolet light (<400 nm). A good correlation was found between the phototoxic effects and known carcinogenic potency (IPCS, 1998).

Sensitisation

Skin Sensitisation

Limited data are available. The chemical, CAS No. 65996-89-6, gave positive results in a single local lymph node assay (LLNA) (REACHa). The polyaromatic hydrocarbon benzo[a]pyrene is classified as hazardous with the risk phrase 'May cause sensitisation by skin contact' (R43) in the HSIS and has been shown to elicit a contact hypersensitivity response in mice and guinea pigs (IPCS, 1998). Overall, the available data support classification for all the chemicals in this group (refer to **Recommendation** section).

An LLNA study (OECD TG 429) was conducted in female BALB/c mice (n = 5/concentration) with high-temperature coal tar (CAS No. 65996-89-6), using a 40 % dimethylacetate/30 % acetone/30 % ethanol (DAE 433) mixture as a vehicle. All test concentrations of 0.3, 3 and 30 % had a stimulation index (SI) over three (7.36, 13.52 and 19.44 respectively). The positive control, dinitrochlorbenzene at a 0.5 % concentration, exhibited an SI of 16.94. The three-fold increase in lymphocyte proliferation (EC3) calculation could not be calculated (REACHa).

Repeated Dose Toxicity

Oral

Limited data are available regarding the non-cancer effects of the chemicals.

In a 94-day and 185-day oral feeding study in B6C3F1 mice (12 animals/sex/dose), with coal tar (CAS No. 8007-45-2), no significant systemic effects were observed with a no observed adverse effect level (NOAEL) of 350 and 460 mg/kg bw/day for female and male mice, respectively, (the highest administered dose) were reported (REACHa).

Dermal

Limited data are available regarding the non-cancer effects of the chemicals. In developmental studies in rats and mice, increased liver and kidney weights were observed in dams, with a lowest observed effect level (LOEL) of 500 mg/kg bw/day (ATSDR, 2002).

Inhalation

Limited data are available regarding the non-cancer effects of the chemicals. Fischer 344 (F344) rats and CD-1 mice were exposed to coal tar aerosol (30, 140 and 690 mg/m^3) for six hours/day, five days/week for 13 weeks. Olfactory epithelium lesions were observed at the highest dose in both species. Effects on haematological parameters, such as decreased red blood cell count and haemoglobin concentrations, were observed in both species, although rats were more sensitive with effects observed at 140 mg/m^3 in rats compared with 690 mg/m^3 in mice. Increased liver and kidney weights and histopathological changes were also observed in rats. Overall, the no observed adverse effect concentration (NOAEC) was established as 30 mg/m^3 in rats and 140 mg/m^3 in mice.

Respiratory symptoms consisting of chronic fibrosing pneumonitis have been observed in long-term studies with coal tar vapours in rats and guinea pigs.

Observation in humans

Respiratory effects, including reduced lung function and pneumoconiosis, has been reported in workers exposed to coal tar fumes, although confounding factors such as cigarette smoking could not be ruled out in a number of these studies.

Genotoxicity

Overall, the data available indicate that the chemicals have genotoxic potential (REACHa; REACHb). The available data support classification (refer to **Recommendation** section). Data for benzo[a]pyrene are considered sufficient to indicate that the chemicals could induce mutations in germ cells. This is consistent with the European adopted harmonised classification for pitch, coal tar, high temperature (CAS no. 65996-93-2) (ECHA, 2011).

Given that note M (refer to **Existing Hazard Classification for Worker Health and Safety** section) for carcinogenicity is considered appropriate for CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 (refer to **Carcinogenicity** section), and the cut-off concentration for mixtures is equivalent for the mutagenicity and carcinogenicity classifications, a similar note for the proposed genotoxicity classification is likely to be appropriate for CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2.

For the chemical high-temperature coal tar (CAS No. 65996-89-6), positive results for genotoxicity were reported in a bacterial reverse mutation assay (Ames test) in *Salmonella typhimurium* strains TA 98, 100, 102 and 1537 with metabolic activation, and ambiguous without metabolic activation. The results were negative for genotoxicity in an Ames test in *S. typhimurium* strain TA 1535 with and without metabolic activation (REACHa).

For the chemical (CAS No. 65996-93-2), positive results for genotoxicity were reported in an Ames test in *S. typhimurium* strains TA 98, 100, and 1537 with metabolic activation, and positive for *S. typhimurium* strains TA 98 without metabolic activation. The results were negative in *S. typhimurium* strains TA 102 and 1535 (REACHb).

Coal-tar pitch emissions have also been reported as mutagenic in mammalian cells with and without metabolic activation, and also to induce sister chromatid exchange (SCE) in Chinese hamster ovary cells and enhanced viral transformation in Syrian hamster embryo cells (IARC, 2012).

DNA adduct formation in mice has been observed with adducts detected in the liver, lungs and forestomach (ATSDR, 2002).

Observation in humans

An association between exposure to coal tar applied dermally as a 2 % or 4 % coal tar ointment and the mutagenicity of urine samples using *S. typhimurium* was reported. An increase in SCE in the peripheral blood lymphocytes of workers exposed to coal tar and coal tar pitches has been reported. DNA adducts have been observed in the blood of a number of patients undergoing dermal coal tar therapy, although consistent results have not been reported. Increased DNA adducts in skin biopsy samples were observed in patients undergoing coal tar therapy with ointments containing 3–5 % and 10 % coal tar for seven or 21 days, respectively. DNA strand breaks have been observed in the peripheral blood of roofers exposed to the dust from coal tar pitch. The available human data suggest genotoxicity (ATSDR, 2002; IARC, 2012).

Genotoxicity of PAHs

The chemicals have the potential to contain several PAHs that are genotoxic including benz[a]anthracene, benzo[b]fluoranthene, benzo[k]fluoranthene, benzo[a]pyrene, dibenz[a,h]anthracene, chrysene, and indeno[1,2,3-cd]pyrene (IARC, 2012). Positive effects were seen in most assays for the mutagenicity of benzo[a]pyrene including inducing sperm abnormalities in mice (IPCS, 1998).

Carcinogenicity

The chemicals (CAS No. 8007-45-2 and CAS No. 65996-89-6) are classified as hazardous—Category 1 carcinogenic substance—with the risk phrase 'May cause cancer' (T; R45) and the chemicals (CAS No. 65996-93-2, CAS No. 68187-57-5, CAS No. 90669-57-1 and CAS No. 90669-58-2) are classified as hazardous—Category 2 carcinogenic substance—with the risk phrase 'May cause cancer' (T; R45) in the HSIS (Safe Work Australia). The available data support the existing Category 1 classification and also support an amendment to the existing Category 2 classification (refer to **Recommendation** section) (IARC, 2012).

Dermal exposure to coal tar and coal tar pitches caused skin tumours in mice. Inhalation exposure to coal tar caused skin tumours in mice and lung tumours in mice and rats (NTP, 2014).

Occupational exposure to coal tars and coal tar pitches is considered to cause skin cancer, scrotal cancer and cancer of the lung. An association between occupational exposure and cancer of the other tissue sites, including the bladder and kidneys, has also been observed (IARC, 1987; IARC; 2012, NTP, 2014).

The International Agency for Research on Cancer (IARC) has classified the chemicals as 'carcinogenic to humans' (Group 1), based on sufficient evidence in humans and experimental animals (IARC, 1987; IARC, 2012).

Coal tars and coal tar pitches are listed by the US National Toxicology Program (NTP) Report on Carcinogens as 'known to be human carcinogens' (NTP, 2014). Coal tar pitch volatiles are classified by the American Conference of Governmental Industrial Hygienists (ACGIH) as an A1 carcinogen (confirmed human carcinogen) (ACGIH, 2011).

No data have been identified regarding the rationale for note M currently applied to CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2. However, this is considered reasonable based on:

- The content of PAHs in coal tars increases as the carbonisation temperature increases; therefore, low temperature coal tar pitches are anticipated to contain lower levels of PAHs.

- Whilst several carcinogenic PAHs might be present as constituents in these chemicals at levels similar or higher than benzo[a]pyrene, the cut-off concentration for mixtures containing category 1 carcinogens is 0.1 % (several orders of magnitude higher than 0.005 %).
- Note M does not currently apply to coal tar and coal tar pitches. Given the levels of PAHs expected, it is not likely that these chemicals would meet the criteria of note M.

Reproductive and Developmental Toxicity

Based the data available, the chemical appears to have potential reproductive or developmental toxicity. Overall, although data on the chemicals themselves are insufficient to warrant classification, data for PAHs, in particular benzo[a]pyrene, are considered sufficient for classification. This is consistent with the European adopted harmonised classification to CAS No. 65996-93-2 (ECHA, 2011).

Note M for carcinogenicity is considered reasonable for CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 (see **Carcinogenicity** section) and the cut-off concentration for mixtures is similar for the reproductive/developmental and carcinogenicity classifications. A similar note for the proposed reproductive/developmental classification is considered appropriate for the chemicals in this group.

The data available on the reproductive effects of the chemicals are not conclusive. Human studies have reported no reproductive hazard from exposure through environmental contamination, or increased risk of spontaneous abortion from coal tar used as a dermal treatment for psoriasis during pregnancy. However, animal studies have shown that exposure through all routes to coal tar (for four or five days during the gestation period) causes increased resorptions in rats and mice. Rats appear more sensitive to these effects than mice. In a 13-week repeated dose inhalation study (see **Repeated dose toxicity**), decreased ovary weights and a decrease in luteal tissue in the ovaries were seen in rats and mice exposed via inhalation to 690 mg/m^3. Increased testicular weights were reported in male rats exposed to 140 mg/m^3 and 690 mg/m^3. This effect was not observed in mice (ATSDR, 2002).

Limited human data were available for developmental effects. However, evidence exists from rat and mice studies that indicates developmental toxicity from exposure to coal tar via all routes. Indications of toxicity include reductions in: foetal ossification, crown-rump length, foetal weight, foetal lung weight and placental weights; significant increases in the incidence of cleft palate; early mortality in pups of treated dams; and prenatal mortality in exposed rat and mouse foetuses. It is noted that the role of maternal toxicity on developmental effects is not clear (ATSDR, 2002).

The chemicals have the potential to contain several PAHs that are embryotoxic including benz[a]anthracene, benzo[a]pyrene, dibenz [a,h]-anthracene, and naphthalene. Benzo[a]pyrene also had adverse effects on female fertility, reproduction and postnatal development (IPCS, 1998).

Risk Characterisation

Critical Health Effects

The critical health effects for risk characterisation are systemic long-term effects including carcinogenicity and reproductive/developmental toxicity. A genotoxic mode of action cannot be ruled out. The chemicals are also considered to be phototoxic and have the potential to cause skin sensitisation and respiratory irritation.

Public Risk Characterisation

Although cosmetic use of coal tars has been identified internationally, coal tars for cosmetic use are currently listed on Appendix C of the SUSMP. The current controls for coal tars are considered adequate to minimise the risk to public health.

Whilst use in driveway sealants has been identified, these are not products used by consumers on a regular basis and, in addition, minimal contact with the chemicals during application is expected. Therefore, the chemicals are not considered to pose an unreasonable risk under the identified uses, and further risk management is not required.

Occupational Risk Characterisation

During product formulation, dermal and inhalation exposure might occur, particularly where manual or open processes are used. These could include transfer and blending activities, quality control analysis, and cleaning and maintaining equipment. Relevant occupations include workers at foundries or those involved in coke production, coal gasification and aluminium production (NTP, 2014).

Worker exposure to the chemicals at lower concentrations could also occur while using formulated products containing the chemicals. The level and route of exposure will vary depending on the method of application and work practices employed. Relevant occupations include workers producing or using pavement tar, roofing tar, coal-tar enamels and other coatings and refractory bricks (NTP, 2014). The potential for road workers and roofers to suffer skin exposure could be considerable as protective clothing might not be worn (NTP, 2014). Using coal tar in road paving has reduced significantly internationally, although workers can continue to be exposed due to recycled coal tar asphalt being used (IARC, 2012). Workers have also been reported to be exposed to significant levels of PAHs while removing coal tar pitch roofs (IARC, 2012).

Given the critical systemic long-term and local health effects, the chemicals could pose an unreasonable risk to workers unless adequate control measures to minimise dermal and inhalation exposure are implemented. The chemicals should be appropriately classified and labelled to ensure that a person conducting a business or undertaking (PCBU) at a workplace (such as an employer) has adequate information to determine the appropriate controls. The prevalence of coal-tar roofs in Australia is unknown. Additional controls might be required to minimise the risk to workers installing and removing coal tar pitch roofs as the potential risk relating to removing existing roofs cannot be controlled through chemical classification and labelling.

Guidance on the *Interpretation of workplace exposure standards for airborne contaminants* advises that exposure to carcinogens should be eliminated or minimised as far as reasonably practicable (Safe Work Australia, 2013).

The data available support an amendment to the hazard classification in the HSIS (Safe Work Australia) (refer to **Recommendation** section).

NICNAS Recommendation

Assessment of these chemicals are considered to be sufficient, provided that the recommended amendment to the classification is adopted, and labelling and all other requirements are met under workplace health and safety and poisons legislation as adopted by the relevant state or territory. It is recommended that Safe Work Australia also consider whether current controls adequately minimise the risk to workers installing and removing coal tar pitch roofs.

Regulatory Control

Work Health and Safety

The chemicals are recommended for classification and labelling under the current approved criteria and adopted GHS as below. This assessment does not consider classification of physical and environmental hazards.

The current HSIS classification for carcinogenicity of the chemicals indicates note H (refer to **Existing Hazard Classification for Worker Health and Safety** section). Note H is no longer considered relevant for these chemicals as the acute, systemic and local effects of the chemicals have been evaluated.

The classifications for CAS Nos. 68187-57-5, 90669-57-1 and 90669-58-2 are subject to note M. Based on this assessment, the genotoxicity and reproductive and developmental classifications provided below should also be subject to note M. Therefore, note M should be slightly modified as follows:

'Note M: The classification (with the exception of classification for sensitisation) need not apply if it can be shown that the substance contains less than 0.005% w/w benzo[a]pyrene (EINECS no. 200-028-5). This note only applies to certain complex coal-derived substances in Annex I.'

Hazard	Approved Criteria (HSIS)[a]	GHS Classification (HCIS)[b]
Sensitisation	May cause sensitisation by skin contact (Xi; R43)	May cause an allergic skin reaction - Cat. 1 (H317)
Genotoxicity	Muta. Cat 2 - May cause heritable genetic damage (T; R46)	May cause genetic defects - Cat. 1B (H340)
Carcinogenicity	Carc. Cat 1 - May cause cancer (T; R45)	May cause cancer - Cat. 1A (H350)
Reproductive and Developmental Toxicity	Repro. Cat 2 - May impair fertility (T; R60) Repro. Cat 2 - May cause harm to the unborn child (T; R61)	May damage fertility or the unborn child - Cat. 1B (H360FD)

[a] Approved Criteria for Classifying Hazardous Substances [NOHSC:1008(2004)].

[b] Globally Harmonized System of Classification and Labelling of Chemicals (GHS) United Nations, 2009. Third Edition.

[*] Existing Hazard Classification. No change recommended to this classification

Advice for consumers

Products containing the chemicals should be used according to the instructions on the label.

Advice for industry

Control measures

Control measures to minimise the risk from dermal and inhalation exposure to the chemicals should be implemented in accordance with the hierarchy of controls. Approaches to minimise risk include substitution, isolation and engineering controls. Measures required to eliminate, or minimise risk arising from storing, handling and using a hazardous chemical depend on the physical form and the manner in which the chemicals are used. Examples of control measures which could minimise the risk include, but are not limited to:

- using closed systems or isolating operations;
- health monitoring for any worker who is at risk of exposure to the chemicals, if valid techniques are available to monitor the effect on the worker’s health;
- air monitoring to ensure control measures in place are working effectively and continue to do so;
- minimising manual processes and work tasks through automating processes;
- work procedures that minimise splashes and spills;
- regularly cleaning equipment and work areas; and
- using protective equipment that is designed, constructed, and operated to ensure that the worker does not come into contact with the chemicals.

Guidance on managing risks from hazardous chemicals are provided in the *Managing risks of hazardous chemicals in the workplace—Code of practice* available on the Safe Work Australia website.

Personal protective equipment should not solely be relied upon to control risk and should only be used when all other reasonably practicable control measures do not eliminate or sufficiently minimise risk. Guidance in selecting personal protective equipment can be obtained from Australian, Australian/New Zealand or other approved standards.

Obligations under workplace health and safety legislation

Information in this report should be taken into account to help meet obligations under workplace health and safety legislation as adopted by the relevant state or territory. This includes, but is not limited to:

- ensuring that hazardous chemicals are correctly classified and labelled;
- ensuring that (material) safety data sheets ((M)SDS) containing accurate information about the hazards (relating to both health hazards and physicochemical (physical) hazards) of the chemicals are prepared; and
- managing risks arising from storing, handling and using a hazardous chemical.

Your work health and safety regulator should be contacted for information on the work health and safety laws in your jurisdiction.

Information on how to prepare an (M)SDS and how to label containers of hazardous chemicals are provided in relevant codes of practice such as the *Preparation of safety data sheets for hazardous chemicals—Code of practice* and *Labelling of workplace hazardous chemicals—Code of practice*, respectively. These codes of practice are available from the Safe Work Australia website.

A review of the physical hazards of these chemicals has not been undertaken as part of this assessment.

References

ACGIH (American Conference of Governmental Industrial Hygienists). Documentation of the Threshold Limit Values for Chemical Substances, ACGIH Signature Publications, 7th Edition, 2011.

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Last Update 13 February 2015

Chemical Identities

Chemical Name in the Inventory and Synonyms	**Tar, coal** coal tar coke-oven tar coking tar crude coal tar aquatar
CAS Number	8007-45-2
Structural Formula	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

Chemical Name in the Inventory and Synonyms	**Tar, coal, high temperature** coal tar coal, high-temp. tar
CAS Number	65996-89-6
Structural Formula	

	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

Chemical Name in the Inventory and Synonyms	**Pitch, coal tar, high temperature** coal tar pitch oil, pitch pitch, coal tar
CAS Number	65996-93-2
Structural Formula	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

Chemical Name in the Inventory and Synonyms	**Pitch, coal tar, petroleum** aromatic pitch, coal petroleum

CAS Number	68187-57-5
Structural Formula	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

Chemical Name in the Inventory and Synonyms	**Pitch, coal tar, low temperature**
CAS Number	90669-57-1
Structural Formula	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

Chemical Name in the	**Pitch, coal tar, low temperature, heat treated**

Inventory and Synonyms	
CAS Number	90669-58-2
Structural Formula	No Structural Diagram Available
Molecular Formula	Unspecified
Molecular Weight	

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